OL | Organic Letters

pubs.acs.org/OrgLett Letter

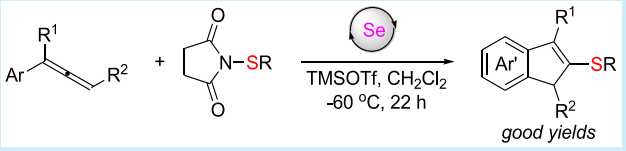
Catalytic Electrophilic Thiocarbocyclization of Allenes

Quanbin Jiang,† Huimin Li,† and Xiaodan Zhao*

Cite This: *Org. Lett.* 2021, 23, 8777–8782 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

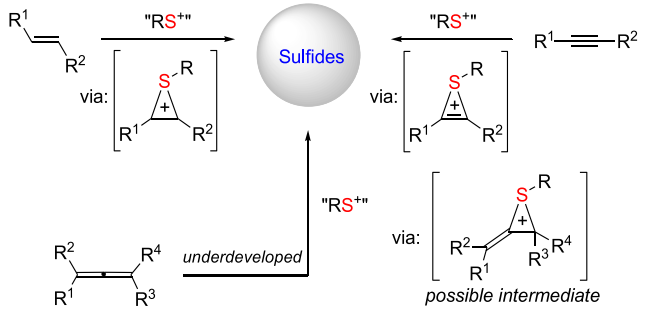
ABSTRACT: An efficient approach via catalytic electrophilic thiocarbocyclization of allenes to construct indene-based sulfides with excellent regioselectivities is disclosed. The reactions were carried out at low temperatures by selenide catalysis in the presence of TMSOTf. Not only electrophilic arylthio reagents but also electrophilic alkylthio reagents worked well under these conditions. Furthermore, the method could be applied to intermolecular azidothiolation of allenes.



Downloaded via FUDAN UNIV on January 17, 2022 at 10:31:21 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Sulfur-containing compounds make up an important class of compounds. Their structural motifs are found in a wide range of biologically active molecules and drugs.[1,2] They can serve as building blocks for the synthesis of valuable compounds and also act as ligands and organocatalysts in organic reactions.[3,4] Due to their importance, numerous methods have been developed to construct these compounds via metal catalysis and organocatalysis in the past several decades.[5,6] For example, transition metal-catalyzed cross-coupling of alkenyl halides with sulfur compounds as well as catalytic hydrothiolation of alkynes to synthesize alkenyl sulfides has been demonstrated.[7] In particular, as a facile route, catalytic electrophilic thiolation of unsaturated hydrocarbon compounds to construct sulfides has attracted a great deal of attention.[5b,8−13] This kind of reaction can be carried out under mild and metal-free conditions. When utilizing electrophilic sulfur reagents, the reactions generally go through a thiiranium or thiirenium ion intermediate, followed by attack of the nucleophiles to give the final sulfide products, which offers a convenient method for quickly constructing multifunctionalized sulfur compounds. In this scenario, the thiolation of alkenes and alkynes has been studied (Scheme 1, top).[8−12] In contrast, the thiolation of allenes, unsaturated compounds with unique structural features, including orthogonal π-orbitals composed of sp^2 and sp carbons, has been rarely investigated (Scheme 1, bottom).[13] To the best of our knowledge, only oxy- and halothiolation of allenes have been demonstrated.

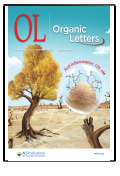
Scheme 1. Synthesis of Sulfides via Electrophilic Thiolation of Carbon−Carbon Unsaturated Compounds



With their unique features, allenes are usually very reactive and can be converted into a variety of products by reacting with different compounds.[14,15] Unlike the electrophilic thiolation of alkenes and alkynes, electrophilic thiolation of allenes might be more challenging for following three reasons. (i) Highly reactive allenes easily decompose under reaction conditions. (ii) Allenes contain two double bonds, resulting in the issue of site selectivity on the three carbons. (iii) The properties of a possible intermediate, thiiranium ion with a vinyl carbon, remain elusive. In spite of the challenges, we discovered that allenes could undergo electrophilic thiocarbocyclization to give sulfides with high regioselectivities by selenide catalysis in the presence of a Lewis acid. Herein, we report the findings.

The Lewis basic selenium-catalyzed electrophilic reaction has emerged as an intriguing strategy for the synthesis of various valuable molecules.[8−10,16,17] In this reaction, Lewis basic selenium compounds can activate electrophilic halogen or sulfur reagents to form a selenium-captured cation ion intermediate. The intermediate has a proper reactivity and can lead to less side products compared with the solo electrophile E^+ ion. This advantage leads to Lewis basic selenium catalysis having good applications in electrophilic halogenation and thiolation of alkenes, alkynes, and even arenes. Our group has been working on organoselenium catalysis. We have developed Lewis basic selenium-catalyzed electrophilic trifluoromethylth-

Received: September 25, 2021
Published: October 28, 2021



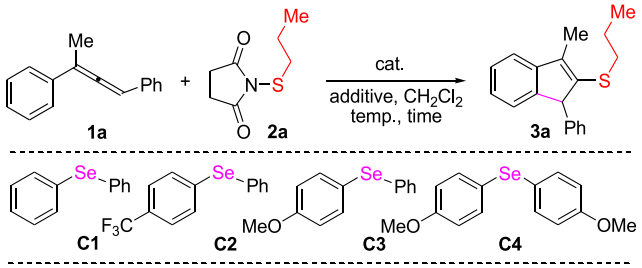
ACS Publications © 2021 American Chemical Society

https://doi.org/10.1021/acs.orglett.1c03270
Org. Lett. 2021, 23, 8777–8782

iolation, long-chain alkylthiolation, and arylthiolation of alkenes.[10a,c−e] Additionally, catalytic electrophilic thiolation of alkynes has been investigated.[10b] As a continuation of our interest, we attempted to extend Lewis basic selenium catalysis to electrophilic thiolation of highly reactive substrates, allenes.

By electrophilic reaction, introducing a long-chain alkylthio group into the parent molecules is more difficult than introducing an arylthio group because of the lower reactivity of electrophilic long-chain alkylthio reagents. To challenge this, we initiated our thiolation study with racemic diphenyl allene **1a** as a model substrate and an electrophilic alkylthiolating reagent, 1-(propylthio)pyrrolidine-2,5-dione (**2a**), as another component. A thiocyclized compound based on a useful indene scaffold was expected as the desired product in the presence of a catalytic amount of selenide (Table 1). When the

Table 1. Optimization of Conditions[a]



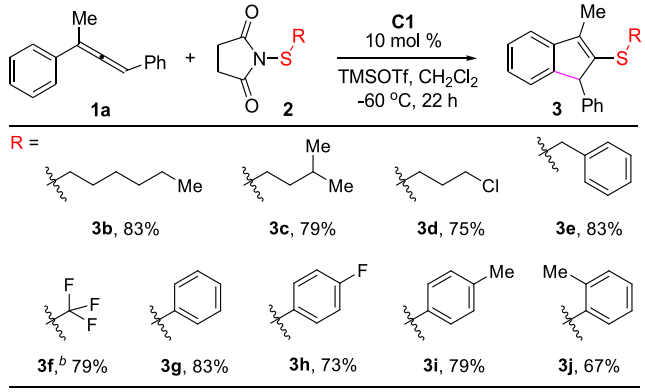
entry	catalyst	additive	temp (°C)	time (h)	yield (%)[b]
1	**C1**	TMSOTf	25	12	0
2	**C1**	TMSOTf	0	12	18
3	**C1**	TMSOTf	−20	12	41
4	**C1**	TMSOTf	−60	12	66
5	**C1**	TMSOTf	−78	12	61
6	**C1**	TMSOTf	−60	22	89 (85)
7	**C1**	TESOTf	−60	22	70
8	**C1**	$BF_3{\cdot}OEt_2$	−60	22	41
9	**C1**	TfOH	−60	22	56
10	**C1**	Tf_2NH	−60	22	55
11[c]	**C1**	TMSOTf	−60	22	63
12[d]	**C1**	TMSOTf	−60	22	69
13	**C2**	TMSOTf	−60	22	70
14	**C3**	TMSOTf	−60	22	67
15	**C4**	TMSOTf	−60	22	68
16	PhSPh	TMSOTf	−60	22	68
17	$Ph_3P{=}S$	TMSOTf	−60	22	49
18	$Ph_3P{=}Se$	TMSOTf	−60	22	18
19	**C1**	–	−60	22	0
20	–	TMSOTf	−60	22	61

[a]Reaction conditions: **1a** (0.05 mmol), **2a** (0.05 mmol), catalyst (10 mol %), additive (0.05 mmol), CH_2Cl_2 (1 mL). [b]NMR yield using 1,3,5-trimethoxybenzene as the internal standard. The isolated yield is in parentheses and on a 0.1 mmol scale. [c]With 0.025 mmol of TMSOTf. [d]With 0.1 mmol of TMSOTf.

reaction was carried out at room temperature using diphenyl selenide as a catalyst and TMSOTf as an additive, no desired product was observed at all (entry 1). Considering the high reactivity of allene **1a**, decreasing the reaction temperature might benefit the reaction. When the reaction was performed at 0 °C, gratifyingly, the desired product was formed in excellent regioselectivity, albeit in low yield (entry 2). Further decreasing the reaction temperature led to higher yields (entries 3 and 4). The reaction gave the best yield at −60 °C (entries 4 and 5). When the reaction time was prolonged to 22 h, the desired product was afforded in 85% isolated yield (entry 6). Other additives such as Lewis acids TESOTf and $BF_3{\cdot}OEt_2$ and Brønsted acids TfOH and Tf_2NH were less effective, leading to lower yields (entries 7−10, respectively). When the amount of additive was reduced to 0.5 equiv or increased to 2 equiv, the products were formed in lower yields (entries 11 and 12). Furthermore, the effect of different chalcogenide catalysts was studied. When using the selenide catalysts bearing electron-withdrawing or electron-donating groups on the phenyl ring, the reactions gave lower yields (entries 13−15). Other chalcogenides such as PhSPh, $Ph_3P{=}S$, and $Ph_3P{=}Se$ had weaker effects on the reaction (entries 16−18, respectively). Possibly, PhSePh provides the optimal Lewis basicity. It is noted that the absence of a Lewis acid resulted in no reaction; however, without a selenide catalyst, the reaction gave the desired product in 61% yield (entries 19 and 20). These results revealed that the Lewis acid plays a significant role in this transformation and the background reaction is strong without selenide.

With the optimal conditions in hand, we first evaluated the scope of electrophilic sulfur reagents (Scheme 2). When the

Scheme 2. Reactions with Different Electrophilic Sulfur Reagents[a]



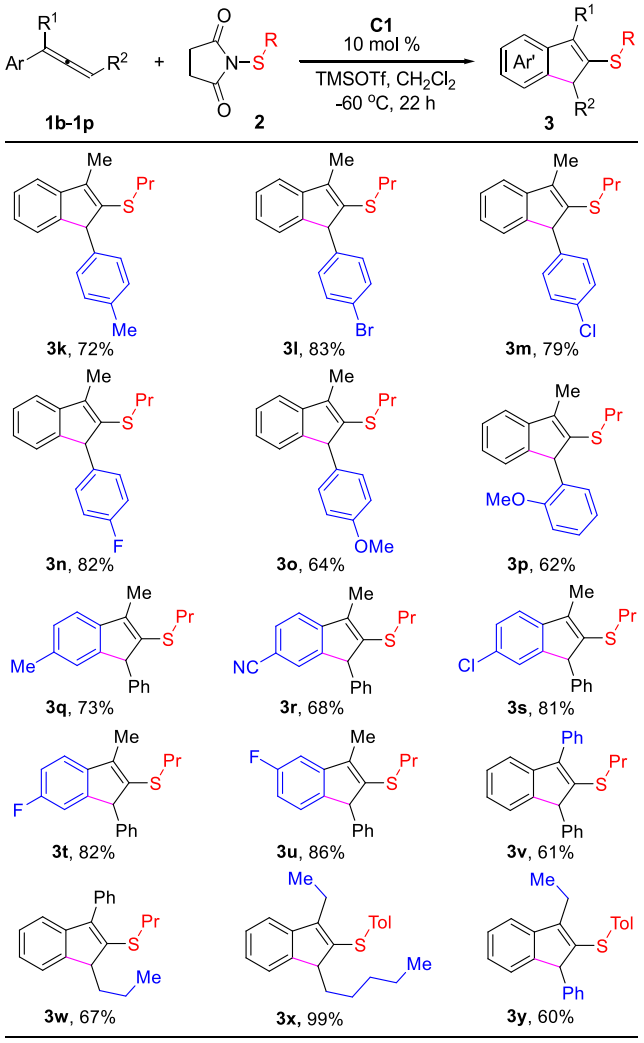
[a]Conditions: **1a** (0.10 mmol), **2** (1.0 equiv), **C1** (10 mol %), TMSOTf (1.0 equiv), CH_2Cl_2 (2.0 mL), −60 °C, 22 h. [b]Using *N*-trifluoromethylthiosaccharin as the sulfur reagent.

alkyl-chain length on the electrophilic alkylthiolating reagents was adjusted, the reactions still proceeded efficiently to give the corresponding products in high yields (**3b**, 83%; **3c**, 79%). Moreover, a benzyl electrophilic sulfur reagent or an alkylthiolating reagent with a halo substituent such as Cl- still afforded the desired products in good yields (**3d**, 75%; **3e**, 83%). Importantly, the method was also suitable for other electrophilic sulfur reagents. Using a more reactive electrophilic trifluoromethylthiolating reagent, the reaction gave the trifluoromethylthio product **3f**, a class of useful compounds,[18] in 79% yield. When arylthiolating reagents were utilized, similarly, the reactions worked well to give the desired products in good yields (**3g**−**3i**, 73−83%). Even if a sterically hindered *o*-methyl-substituted sulfur reagent was used, the desired product **3j** was formed in good yield. These results indicated that the method is general.

Next, the scope of allene substrates were evaluated (Scheme 3). The effect of different substituents on the allene substrates was investigated. When R^1 was a methyl group and R^2 was an

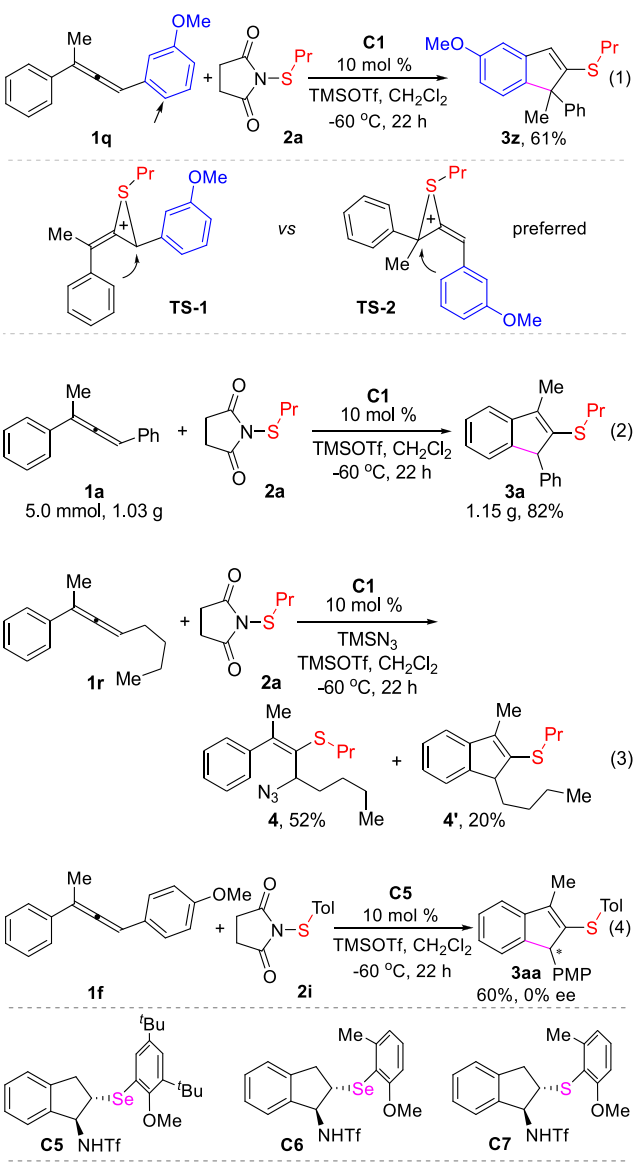
https://doi.org/10.1021/acs.orglett.1c03270
Org. Lett. 2021, 23, 8777−8782

Scheme 3. Reactions with Different Allenes[a]



[a]Conditions: substrates **1b−1p** (0.10 mmol), **2** (1.0 equiv), PhSePh (10 mol %), TMSOTf (1.0 equiv), CH_2Cl_2 (2.0 mL), −60 °C, 22 h.

aryl group, it was found that various substituents on the phenyl group of R^2 had an impact on the reactions. Electron-donating groups resulted in relatively lower yields than electron-withdrawing ones (**3k−3o**, 64−83%). It is noted that when the substituent, e.g., MeO-, was placed at the *ortho* position of the phenyl group, the reaction was slightly affected, still giving the desired product **3p** in good yield. For the aryl nucleophile, the substituents on it had a similar electronic effect compared with the effect of those on the phenyl group of R^2. Electron-donating groups usually led to yields that were relatively lower than those of electron-withdrawing ones except the -CN group (**3q−3u**, 68−86%). Possibly, the -CN group was too electron-withdrawing, so that the corresponding allene was more highly reactive, increasing the likelihood of side reactions. When R^1 was an aryl group and R^2 was an aryl or alkyl group, the reactions still proceeded smoothly to afford the desired products in good yields (**3v**, 61%; **3w**, 67%). Furthermore, the R^1 group could be a long-chain alkyl group. To more easily purify the desired products, an electrophilic arylthiolating reagent was used (**3x**, 99%; **3y**, 60%). Interestingly, when allene **1q** with a methoxy group at the *meta* position was utilized as the substrate, the reaction gave product **3z** in good yield, in which the electron-rich aryl group acted as a nucleophile (eq 1). As depicted in transition states **TS-1** and



TS-2, the methoxy substituent can cause the phenyl group to be more electron-rich. Subsequently, the electron-rich aryl group is preferred as a nucleophile to the phenyl group even in the presence of greater steric hindrance. These results indicated that appropriate electron-rich aryl groups connected to the double bonds can switch the regioselectivity of reactions.

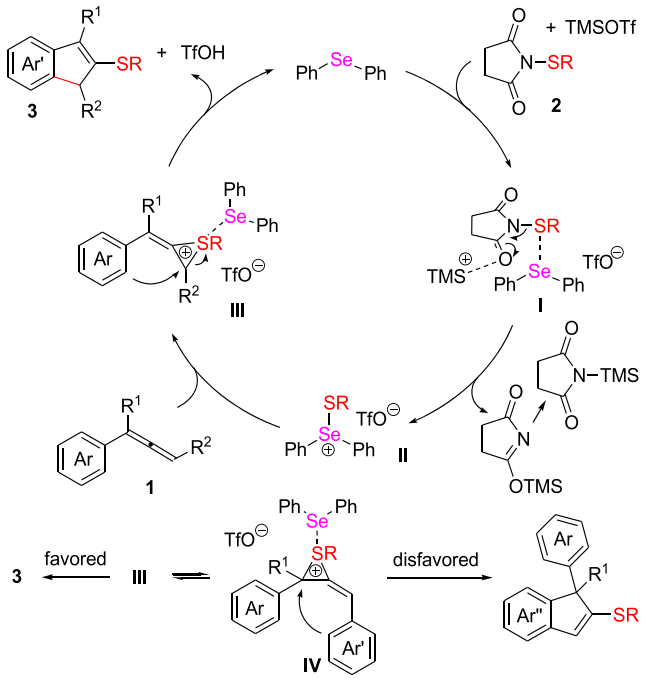
To further explore the efficiency, practicality, and generality of this method, relevant studies were conducted. First, a gram-scale experiment was performed. When **1a** (5.0 mmol) was treated with electrophilic sulfur reagent under similar conditions, product **3a** was still obtained in high yield (eq 2). Then, an intermolecular azidothiolation reaction with **1r** as the substrate was tested under similar conditions (eq 3). The desired product was formed with high regioselectivity in moderate yield along with the formation of a thiocarbocyclization byproduct. This preliminary result reflected the possibility of intermolecular reaction. Finally, the kinetic resolution of this kind of reaction was investigated, although the background

https://doi.org/10.1021/acs.orglett.1c03270
Org. Lett. 2021, 23, 8777−8782

reaction might be strong.[19] When racemic allene **1f** was treated with sulfur reagent **2i** in the presence of chiral bifunctional selenide catalyst **C5**, cyclized product **3aa** could be formed, but without ee. Similarly, the other catalysts such as selenide **C6** and sulfide **C7** delivered no enantioselectivity. Even if the reaction of **1f** ran for 1 h, **3aa** was also formed in 40% yield without ee. The results revealed that the current conditions are not effective for the resolution of racemic allenes.

On the basis of the results presented above and previous studies,[10a] a possible mechanism is proposed in Scheme 4. An

Scheme 4. Proposed Mechanism



electrophilic sulfur reagent is first activated by the PhSePh catalyst with the assistance of Lewis acid TMSOTf via transition state **I**. A selenide-captured sulfonium ion intermediate **II** is formed. Then, reaction of **II** with substrates gives cyclic thiiranium ion intermediate **III** with high regioselectivity. Subsequent nucleophilic attack of the aromatic ring on the thiiranium ion unit affords the desired products **3** and regenerates the selenide catalyst. However, the properties of the cyclic thiiranium ion with a vinyl cabon are unknown. The thiiranium ion might be very unstable to decomposition to the corresponding carbon cation and then undergo nucleophilic attack to give the final products. This could be one of the potential reasons why the kinetic resolution of the reaction does not work. Presumably, the thiiranium ion intermediate, e.g., **IV**, formed from the other double bond on the allene substrate might be more stable. Due to the steric hindrance that disfavors the formation of the quaternary carbon center, the subsequent cyclization may not proceed easily. Only in the case of **3z** does the strong nucleophilicity of the electron-rich aryl group overcome the steric hindrance to furnish product **3z**. There might be an equilibrium between intermediates **IV** and **III**, leading to products **3**.

In summary, we have developed an efficient method for the synthesis of indene-based sulfides with high regioselectivities in good yields via organoselenium-catalyzed electrophilic thiocarbocyclization of allenes. This method could be applied to a similar intermolecular reaction. It is noted that the reaction temperature is crucial in the reactions. This is the first example of Lewis basic chalcogenide-catalyzed reaction of allenes. Further investigations of chalcogenide-catalyzed reactions are ongoing in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c03270.

Experimental details, characterization data, and NMR spectra of new compounds (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Xiaodan Zhao − *Institute of Organic Chemistry & MOE Key Laboratory of Bioinorganic and Synthetic Chemistry, School of Chemistry, Sun Yat-Sen University, Guangzhou 510275, P. R. China;* orcid.org/0000-0002-2135-5121; Email: zhaoxd3@mail.sysu.edu.cn

Authors

Quanbin Jiang − *Institute of Organic Chemistry & MOE Key Laboratory of Bioinorganic and Synthetic Chemistry, School of Chemistry, Sun Yat-Sen University, Guangzhou 510275, P. R. China;* orcid.org/0000-0002-0321-2688

Huimin Li − *Institute of Organic Chemistry & MOE Key Laboratory of Bioinorganic and Synthetic Chemistry, School of Chemistry, Sun Yat-Sen University, Guangzhou 510275, P. R. China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.1c03270

Author Contributions

†Q.J. and H.L. contributed equally to this work.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors thank the National Natural Science Foundation of China (Grants 21901261, 21772239, and 91856109), the Fundamental Research Funds for the Central Universities (Grant 20lgzd21), and the Leading Scientific, Technical and Innovation Talents of Guangdong Special Support Program (2019TX05Y638) for financial support.

■ REFERENCES

(1) (a) Ilardi, E. A.; Vitaku, E.; Njardarson, J. T. Data-Mining for Sulfur and Fluorine: An Evaluation of Pharmaceuticals To Reveal Opportunities for Drug Design and Discovery. *J. Med. Chem.* **2014**, *57*, 2832−2842. (b) Feng, M.; Tang, B.; Liang, S. H.; Jiang, X. Sulfur Containing Scaffolds in Drugs: Synthesis and Application in Medicinal Chemistry. *Curr. Top. Med. Chem.* **2016**, *16*, 1200−1216.

(2) (a) Anne, C.; Turcaud, S.; Quancard, J.; Teffo, F.; Meudal, H.; Fournie-Zaluski, M.-C.; Roques, B. P. Development of Potent Inhibitors of Botulinum Neurotoxin Type B. *J. Med. Chem.* **2003**, *46*, 4648−4656. (b) McGrath, N. A.; Brichacek, M.; Njardarson, J. T. A Graphical Journey of Innovative Organic Architectures That Have Improved Our Live. *J. Chem. Educ.* **2010**, *87*, 1348−1349. (c) Scott, K. A.; Njardarson, J. T. Analysis of US FDA-Approved Drugs Containing Sulfur Atoms. *Top. Curr. Chem.* **2018**, *376*, 5.

(3) Chatgilialoglu, C., Asmus, K. D., Eds. *Sulfur-Centered Reactive Intermediates in Chemistry and Biology*; Springer Verlag: New York, 1991.
(4) (a) Masdeu-Bultó, A. M.; Diéguez, M.; Martin, E.; Gómez, M. Chiral thioether ligands: coordination chemistry and asymmetric catalysis. *Coord. Chem. Rev.* **2003**, *242*, 159−201. (b) Otocka, S.; Kwiatkowska, M.; Madalińska, L.; Kiełbasiński, P. Chiral Organosulfur Ligands/Catalysts with a Stereogenic Sulfur Atom: Applications in Asymmetric Synthesis. *Chem. Rev.* **2017**, *117*, 4147−4181.
(5) (a) Kondo, T.; Mitsudo, T.-a. Metal-Catalyzed Carbon-Sulfur Bond Formation. *Chem. Rev.* **2000**, *100*, 3205−3220. (b) Chauhan, P.; Mahajan, S.; Enders, D. Organocatalytic Carbon−Sulfur Bond-Forming Reactions. *Chem. Rev.* **2014**, *114*, 8807−8864. (c) Lee, C.-F.; Liu, Y.-C.; Badsara, S. S. Transition-Metal-Catalyzed C-S Bond Coupling Reaction. *Chem. - Asian J.* **2014**, *9*, 706−722. (d) Yu, J.-S.; Huang, H.-M.; Ding, P.-G.; Hu, X.-S.; Zhou, F.; Zhou, J. Catalytic Enantioselective Construction of Sulfur-Containing Tetrasubstituted Carbon Stereocenters. *ACS Catal.* **2016**, *6*, 5319−5344.
(6) For selected examples, see: (a) Pritzius, A. B.; Breit, B. Asymmetric Rhodium-Catalyzed Addition of Thiols to Allenes: Synthesis of Branched Allylic Thioethers and Sulfones. *Angew. Chem., Int. Ed.* **2015**, *54*, 3121−3125. (b) Kennemur, J. L.; Kortman, G. D.; Hull, K. L. Rhodium-Catalyzed Regiodivergent Hydrothiolation of Allyl Amines and Imines. *J. Am. Chem. Soc.* **2016**, *138*, 11914−11919. (c) Yang, X.-H.; Davison, R.; Nie, S.-Z.; Cruz, F. A.; McGinnis, T. M.; Dong, V. M. Catalytic Hydrothiolation: Counterion-Controlled Regioselectivity. *J. Am. Chem. Soc.* **2019**, *141*, 3006−3013. (d) Nie, S.-Z.; Lu, A.; Kuker, E. L.; Dong, V. M. Enantioselective Hydrothiolation: Diverging Cyclopropenes through Ligand Control. *J. Am. Chem. Soc.* **2021**, *143*, 6176−6184.
(7) (a) Weiss, C. J.; Marks, T. J. Organozirconium Complexes as Catalysts for Markovnikov-Selective Intermolecular Hydrothiolation of Terminal Alkynes: Scope and Mechanism. *J. Am. Chem. Soc.* **2010**, *132*, 10533−10546. (b) Di Giuseppe, A.; Castarlenas, R.; Perez-Torrente, J. J.; Crucianelli, M.; Polo, V.; Sancho, R.; Lahoz, F. J.; Oro, L. A. Ligand-Controlled Regioselectivity in the Hydrothiolation of Alkynes by Rhodium N-Heterocyclic Carbene Catalysts. *J. Am. Chem. Soc.* **2012**, *134*, 8171−8183. (c) Dondoni, A.; Marra, A. Metal-Catalyzed and Metal-Free Alkyne Hydrothiolation: Synthetic Aspects and Application Trends. *Eur. J. Org. Chem.* **2014**, *2014*, 3955−3969. (d) Strydom, I.; Guisado-Barrios, G.; Fernandez, I.; Liles, D. C.; Peris, E.; Bezuidenhout, D. I. A Hemilabile and Cooperative N-Donor-Functionalized 1,2,3-Triazol-5-Ylidene Ligand for Alkyne Hydrothiolation Reactions. *Chem. - Eur. J.* **2017**, *23*, 1393−1401. (e) Velasco, N.; Virumbrales, C.; Sanz, R.; Suarez-Pantiga, S.; Fernandez-Rodriguez, M. A. General Synthesis of Alkenyl Sulfides by Palladium-Catalyzed Thioetherification of Alkenyl Halides and Tosylates. *Org. Lett.* **2018**, *20*, 2848−2852.
(8) Matviitsuk, A.; Panger, J. L.; Denmark, S. E. Catalytic, Enantioselective Sulfenofunctionalization of Alkenes: Development and Recent Advances. *Angew. Chem., Int. Ed.* **2020**, *59*, 19796−19819.
(9) (a) Luo, J.; Liu, X.; Zhao, X. Development of Chalcogenide Catalysts towards Trifluoromethylthiolation. *Synlett* **2017**, *28*, 397−401. (b) Jiang, Q.; Zhao, X. Chiral Bifunctional Chalcogenide-Catalyzed Enantioselective Electrophilic Thiofunctionalization of Alkenes. *Chin. J. Org. Chem.* **2021**, *41*, 443−454.
(10) (a) Luo, J.; Zhu, Z.; Liu, Y.; Zhao, X. Diaryl Selenide Catalyzed Vicinal Trifluoromethylthioamination of Alkenes. *Org. Lett.* **2015**, *17*, 3620−3623. (b) Wu, J.-J.; Xu, J.; Zhao, X. Selenide-Catalyzed Stereoselective Construction of Tetrasubstituted Trifluoromethylthiolated Alkenes with Alkynes. *Chem. - Eur. J.* **2016**, *22*, 15265−15269. (c) Liu, X.; Liang, Y.; Ji, J.; Luo, J.; Zhao, X. Chiral Selenide-Catalyzed Enantioselective Allylic Reaction and Intermolecular Difunctionalization of Alkenes: Efficient Construction of C-SCF_3 Stereogenic Molecules. *J. Am. Chem. Soc.* **2018**, *140*, 4782−4786. (d) Liang, Y.; Zhao, X. Enantioselective Construction of Chiral Sulfides via Catalytic Electrophilic Azidothiolation and Oxythiolation of *N*-Allyl Sulfonamides. *ACS Catal.* **2019**, *9*, 6896−6902. (e) Zhang, Y.; Liang, Y.; Zhao, X. Chiral Selenide-Catalyzed, Highly Regio- and Enantioselective Intermolecular Thioarylation of Alkenes with Phenols. *ACS Catal.* **2021**, *11*, 3755−3761.
(11) (a) Guan, H.; Wang, H.; Huang, D.; Shi, Y. Enantioselective oxysulfenylation and oxyselenenylation of olefins catalyzed by chiral Brønsted acids. *Tetrahedron* **2012**, *68*, 2728−2735. (b) Li, L.; Li, Z.; Huang, D.; Wang, H.; Shi, Y. Chiral phosphoric acid catalyzed enantioselective sulfamination of amino-alkenes. *RSC Adv.* **2013**, *3*, 4523−4525. (c) Luo, H.-Y.; Xie, Y.-Y.; Song, X.-F.; Dong, J.-W.; Zhu, D.; Chen, Z.-M. Lewis base-catalyzed asymmetric sulfenylation of alkenes: construction of sulfenylated lactones and application to the formal syntheses of (−)-nicotlactone B and (−)-galbacin. *Chem. Commun.* **2019**, *55*, 9367−9370. (d) Xie, Y.-Y.; Chen, Z.-M.; Luo, H.-Y.; Shao, H.; Tu, Y.-Q.; Bao, X.; Cao, R.-F.; Zhang, S.-Y.; Tian, J.-M. Lewis Base/Brønsted Acid Co-catalyzed Enantioselective Sulfenylation/Semipinacol Rearrangement of Di- and Trisubstituted Allylic Alcohols. *Angew. Chem., Int. Ed.* **2019**, *58*, 12491−12496.
(12) (a) Hu, B.-L.; Pi, S.-S.; Qian, P.-C.; Li, J.-H.; Zhang, X.-G. Palladium-Catalyzed Iodine-Mediated Electrophilic Annulation of 2-(1-Alkynyl)biphenyls with Disulfides. *J. Org. Chem.* **2013**, *78*, 1300−1305. (b) Li, X.; Guo, Y.; Shen, Z. Hydrochloric Acid-Promoted Intermolecular 1,2-Thiofunctionalization of Aromatic Alkenes. *J. Org. Chem.* **2018**, *83*, 2818−2829. (c) Zhang, Z.; He, P.; Du, H.; Xu, J.; Li, P. Sulfur-Mediated Electrophilic Cyclization of Aryl-Substituted Internal Alkynes. *J. Org. Chem.* **2019**, *84*, 4517−4524.
(13) (a) Jacobs, T. L.; Macomber, R. S.; Zunker, D. Addition reactions of allenes. III. 2,4-Dinitrobenzenesulfenyl chloride and bromine. *J. Am. Chem. Soc.* **1967**, *89*, 7001−7005. (b) Capozzi, G.; Franck, R. W.; Mattioli, M.; Menichetti, S.; Nativi, C.; Valle, G. Phthalimidesulfenyl Chloride. 9. A Simple Access to.alpha.alpha.'-Dioxothiones, a New Class of Bis-heterodienes. Synthesis of 1,4-Oxathiin Systems. *J. Org. Chem.* **1995**, *60*, 6416−6426. (c) Christov, V. C.; Ivanov, I. K.; Ismailov, I. E. Bifunctionalized Allenes. Part X. An Electrophilic Cyclization Protocol for Convenient Highly Regioselective Synthesis of 3-Sulfonylfuran-2(5*H*)-ones from 2-Sulfonylallenoates. *Heterocycles* **2013**, *87*, 1903−1916. (d) Hasanov, H. H.; Ivanov, I. K.; Christov, V. C. Bifunctionalized allenes. Part XXI. Electrophilic cyclization and addition reactions of 3-(α- or β-hydroxyalkyl)-allenylphosphonates and allenyl phosphine oxides. *Phosphorus, Sulfur Silicon Relat. Elem.* **2018**, *193*, 611−619. (e) Yan, J.; Jiang, M.; Song, L.-P.; Liu, J.-T. Oxytrifluoromethylthiolation of 2,3-Allenoates with Trifluoromethanesulfinyl Chloride: A Synthetic Approach to Trifluoromethylthiolated 4-Oxo-2(E)-alkenoates and Furans. *Adv. Synth. Catal.* **2020**, *362*, 2882−2887.
(14) For recent reviews, see: (a) Le Bras, J.; Muzart, J. Palladium-catalysed inter- and intramolecular formation of C−O bonds from allenes. *Chem. Soc. Rev.* **2014**, *43*, 3003−3040. (b) Wang, Z.; Xu, X.; Kwon, O. Phosphine catalysis of allenes with electrophiles. *Chem. Soc. Rev.* **2014**, *43*, 2927−2940. (c) Ye, J.; Ma, S. Palladium-Catalyzed Cyclization Reactions of Allenes in the Presence of Unsaturated Carbon−Carbon Bonds. *Acc. Chem. Res.* **2014**, *47*, 989−1000. (d) Lledo, A.; Pla-Quintana, A.; Roglans, A. Allenes, versatile unsaturated motifs in transition-metal-catalysed [2+2+2] cycloaddition reactions. *Chem. Soc. Rev.* **2016**, *45*, 2010−2023. (e) Yang, B.; Qiu, Y.; Backvall, J.-E. Control of Selectivity in Palladium(II)-Catalyzed Oxidative Transformations of Allenes. *Acc. Chem. Res.* **2018**, *51*, 1520−1531. (f) Liu, L.; Ward, R. M.; Schomaker, J. M. Mechanistic Aspects and Synthetic Applications of Radical Additions to Allenes. *Chem. Rev.* **2019**, *119*, 12422−12490. (g) Blieck, R.; Taillefer, M.; Monnier, F. Metal-Catalyzed Intermolecular Hydrofunctionalization of Allenes: Easy Access to Allylic Structures via the Selective Formation of C−N, C−C, and C−O Bonds. *Chem. Rev.* **2020**, *120*, 13545−13598.
(15) For selected examples, see: (a) Abdul-Malik, N.; Awad, S. B.; Sakla, A. B. Halogenation of Xanthenyl and Thioxanthenylallenes. *Helv. Chim. Acta* **1979**, *62*, 1872−1877. (b) Bartels, H. M.; Boldt, P. Radikalische Addition von Brommalononitrilan Allene. *Liebigs Ann. Chem.* **1981**, *1981*, 40−46. (c) Kong, W.; Guo, B.; Fu, C.; Ma, S. An Efficient Approach to 2-Bromoalken-3-ols by Regioselective Bromohydroxylation Reaction of Simple Allenes with NBS. *Eur. J. Org. Chem.*

https://doi.org/10.1021/acs.orglett.1c03270
Org. Lett. 2021, 23, 8777−8782

2011, *2011*, 2278−2285. (d) Grandclaudon, C.; Michelet, V.; Toullec, P. Y. Synthesis of Polysubstituted 2-Iodoindenes via Iodonium-Induced Cyclization of Arylallenes. *Org. Lett.* **2016**, *18*, 676−679. (e) Song, E.; Kim, H. Y.; Oh, K. A facile synthesis of 3,4-dimercaptofurans *via* sulfenylation of (*E*)-*β*-chlorovinyl ketones and 1,2-sulfur migration. *Org. Biomol. Chem.* **2017**, *15*, 1776−1779. (f) Goh, J.; Maraswami, M.; Loh, T.-P. Synthesis of Vinylic Sulfones in Aqueous Media. *Org. Lett.* **2021**, *23*, 1060−1065.

(16) Chen, F.; Tan, C. K.; Yeung, Y.-Y. C_2-Symmetric Cyclic Selenium-Catalyzed Enantioselective Bromoaminocyclization. *J. Am. Chem. Soc.* **2013**, *135*, 1232−1235.

(17) Guo, R.; Liu, Z.; Zhao, X. Efficient Synthesis of P-Chirogenic Compounds Enabled by Chiral Selenide-Catalyzed Enantioselective Electrophilic Aromatic Halogenation. *CCS Chem.* **2020**, *2*, 2617−2628.

(18) (a) Leroux, F.; Jeschke, P.; Schlosser, M. *α*-Fluorinated Ethers, Thioethers, and Amines: Anomerically Biased Species. *Chem. Rev.* **2005**, *105*, 827−856. (b) Muller, K.; Faeh, C.; Diederich, F. Fluorine in Pharmaceuticals: Looking Beyond Intuition. *Science* **2007**, *317*, 1881−1886. (c) Purser, S.; Moore, P. R.; Swallow, S.; Gouverneur, V. Fluorine in medicinal chemistry. *Chem. Soc. Rev.* **2008**, *37*, 320−330.

(19) Denmark, S. E.; Chi, H. M. Catalytic, Enantioselective, Intramolecular Carbosulfenylation of Olefins. Mechanistic Aspects: A Remarkable Case of Negative Catalysis. *J. Am. Chem. Soc.* **2014**, *136*, 3655−3663.

https://doi.org/10.1021/acs.orglett.1c03270
Org. Lett. 2021, 23, 8777−8782